Causal Random Forests

From potential outcomes to honest causal forests

Eugeni Gil-Ocana

Universitat Politècnica de València

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Outline

The fundamental problem of causal inference

- Supervised ML learns $\mathbb{E}[Y \mid X, W]$: what *is*, not what *would be*.
- Causal questions are inherently counterfactual: "what would happen if we intervened?"
- We can never observe the same unit under both treatment and control.
- Observational data conflates selection bias with the true causal effect.

Holland (1986) called this *the Fundamental Problem of Causal Inference*: it is impossible to observe the effect of treatment t on unit i, because we can never see $Y_i(1)$ and $Y_i(0)$ simultaneously.

Why OLS isn't enough

Suppose $W \in \{0,1\}$ is a binary treatment and the true outcome model is $Y = \tau W + \beta_U U + \varepsilon$, with U an unobserved confounder. A naive regression of Y on W gives:

$$\hat{\tau}_{\mathrm{OLS}} \xrightarrow{p} \tau + \beta_U \cdot \underbrace{\frac{\operatorname{Cov}(W, U)}{\operatorname{Var}(W)}}_{\text{omitted-variable bias}} \neq \tau$$

Adding observed controls X only helps if X fully captures the confounding; any remaining U leaves the same structure of bias, with covariances taken *partial* on X.

The Rubin causal model

For each unit $i = 1, \ldots, n$:

Potential outcomes. Let $Y_i(1)$ be the outcome if treated and $Y_i(0)$ the outcome if not. Only one is ever observed:

$$Y_i^{\text{obs}} = W_i \cdot Y_i(1) + (1 - W_i) \cdot Y_i(0)$$

where $W_i \in \{0, 1\}$ is the received treatment.

We want the individual treatment effect $Y_i(1) - Y_i(0)$ but it is *never observed for any single unit*. Yet although individual effects remain hidden, their expectation is estimable: averages survive even when individual realizations do not. The entire field is built on recovering these expectations from observational data.

The three identification assumptions

1. SUTVA = Stable Unit Treatment Value: no interference between units, no hidden versions of treatment.
2. Unconfoundedness = conditional independence:

$$\{Y_i(0),\ Y_i(1)\} \perp\!\!\!\perp W_i \,\Big|\, X_i$$

   Once we condition on X_i, the data behave *as if* they came from a randomized experiment.
3. Overlap = positivity:

$$0 \ < \ e(x) \ < \ 1 \quad \forall x \in \mathrm{supp}(X)$$

   where $e(x) = \mathbb{P}(W_i = 1 \mid X_i = x)$ is the propensity score.

SUTVA and unconfoundedness are untestable. Overlap is the exception, we can check it from the data.

Identification result

Under SUTVA, unconfoundedness, and overlap, the conditional average treatment effect (CATE) is non-parametrically identified from observational data:

$$\underbrace{\tau(x) := \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]}_{\text{counterfactual — unobservable}} = \underbrace{\mathbb{E}[Y_i \mid X_i = x, W_i = 1] - \mathbb{E}[Y_i \mid X_i = x, W_i = 0]}_{\text{observable from data}}$$

This is the magic moment of causal inference: a counterfactual quantity equals a difference of conditional expectations that we can estimate. Everything that follows is about doing this estimation *well* when X is high-dimensional.

The natural (wrong) idea

Fit two separate outcome models and subtract the T-learner ("T" for "two models"):

$$\mu_w(x) = \mathbb{E}[Y \mid X = x,\ W = w], \qquad \hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$$

What's wrong with it?

The double-burden problem. Each model must (i) control for confounding *and* (ii) detect how the effect varies with X.

Example: if wages grow steeply with age (large baseline) and the program adds a modest 5%, both models must learn the age-wage curve *perfectly* to recover the 5% signal. Any error in the baseline contaminates $\hat{\tau}$.

The model behind everything

Assume the effect is additive but the baseline is arbitrary, the *partially linear model*:

$$Y_i \;=\; \tau(X_i)\, W_i \;+\; g(X_i) \;+\; \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid X_i, W_i] = 0.$$

We never want to estimate the baseline g, only τ. Take $\mathbb{E}[\,\cdot \mid X_i]$ of both sides and two objects appear on their own:

$$m(x) = \mathbb{E}[Y_i \mid X_i = x] \;=\; \tau(x)\, e(x) + g(x) \qquad \text{outcome mean}$$
$$e(x) = \mathbb{E}[W_i \mid X_i = x] \;=\; \mathbb{P}(W_i = 1 \mid X_i = x) \qquad \text{propensity}$$

Crucial: $m(x) \neq \mu_0(x)$. The mean m averages over treated *and* control, so it never looks at W on its own it is the treatment-blind prediction of Y.

The Robinson decomposition

Subtract the m-line from the original model. The baseline $g(X_i)$ is in both, so it cancels:

$$\underbrace{Y_i - m(X_i)}_{\tilde{Y}_i} = \tau(X_i)\left(\underbrace{W_i - e(X_i)}_{\tilde{W}_i}\right) + \varepsilon_i.$$

ROBINSON (1988) — ECONOMETRICA

$$\tilde{Y}_i = \tau(X_i)\,\tilde{W}_i + \varepsilon_i$$

Remove what X predicts about Y and about W.
What remains is the treatment variation used for causal estimation
under the identification assumptions.

The nuisance g never reappears, we only ever estimate m and e. No intercept: the slope **is** the effect.
And $\mathbb{E}[\tilde{W}_i \mid X_i] = 0$ by construction, the fact that makes the next slide work.

Double Machine Learning

We never know m, e exactly. Do estimation errors leak into τ? Not at first order: an error δ_m in $\hat{m}$ enters multiplied by $\tilde{W}$, which is mean-zero given X, so it cancels:

$$\mathbb{E}\left[\delta_m(X_i)\,\tilde{W}_i \mid X_i\right] = \delta_m(X_i) \cdot \underbrace{\mathbb{E}[\tilde{W}_i \mid X_i]}_{=0} = 0.$$

Only the *product* of the two nuisance errors survives: $\mathrm{Bias}[\hat{\tau}] = O\big(\|\hat{m} - m\|_2 \cdot \|\hat{e} - e\|_2\big)$.

Thus neither nuisance must be estimated at the parametric $n^{-1/2}$ rate. A standard sufficient condition is that both nuisances are $o(n^{-1/4})$ in suitable L_2 error, plausible for flexible ML under regularity, sparsity, or smoothness conditions.

This is Double Machine Learning (Chernozhukov et al. 2018): orthogonal score + cross-fitting $\Rightarrow$ valid inference with ML nuisances. The residual-on-residual regression is its leading example.

The recipe: three steps before the forest

The T-learner estimated τ from the noisy baseline; we estimate it from the *residuals* instead — in three steps:

1. Estimate the nuisances. Any flexible learner (random forest, boosting, neural net) for $\hat{m}(x)$ and $\hat{e}(x)$ — out-of-sample (cross-fitting).
2. Compute residuals. $\tilde{Y}_i = Y_i - \hat{m}(X_i), \quad \tilde{W}_i = W_i - \hat{e}(X_i)$.
3. Regress $\tilde{Y}$ on $\tilde{W}$. The coefficient is $\tau(X_i)$. *This step is where the Causal Random Forest enters*, it runs this regression *locally*, with adaptive weights.

Frisch–Waugh–Lovell as a special case. If $\tau(x) = \tau$ (homogeneous), step 3 is just OLS of $\tilde{Y}$ on $\tilde{W}$: $\hat{\tau} = \sum_i \tilde{W}_i \tilde{Y}_i / \sum_i \tilde{W}_i^2$.

The R-learner: Robinson as a loss function

Nie & Wager (2021) noticed that Robinson defines a natural loss for any function class:

$$\hat{\tau} \;=\; \arg\min_{\tau \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \big(\tilde{Y}_i - \tau(X_i) \cdot \tilde{W}_i\big)^2 \;+\; \lambda \, \|\tau\|^2$$

"R" stands for Robinson *and* for residual. Choose your function class:

Method	$\hat{\tau}$ **from** $\tilde{Y}, \tilde{W}$
R-learner (linear)	Ridge/Lasso of $\tilde{Y}$ on $\tilde{W} \cdot X$
R-learner (boosting)	XGBoost with weights $\tilde{W}_i^2$, target $\tilde{Y}_i / \tilde{W}_i$
Causal Forest (econml)	Locally weighted OLS of $\tilde{Y}$ on $\tilde{W}$, forest weights $\alpha_i(x)$
R-learner (neural net)	Minimise the R-loss with τ a neural net

The regression tree

Before the causal version, recall what a tree does for a *conditional mean*. We want $\mu(x) \equiv \mathbb{E}[Y \mid X = x]$. Let $L(x, T)$ be the leaf of tree T that contains x. The estimator is just the leaf average:

$$\hat{\mu}(x) = \frac{1}{\left|\{i : X_i \in L(x, T)\}\right|} \sum_{i : X_i \in L(x, T)} Y_i.$$

The tree itself is grown (greedily) to minimise the in-sample squared error:

$$\sum_{i=1}^{n} \left(Y_i - \hat{\mu}(X_i)\right)^2.$$

The split rule and the leaf value both target the *same* observable, Y. The causal version breaks exactly here because the object of interest is never observed.

From regression to causal trees

Now change the target: the CATE $\tau(x) \equiv \mathbb{E}[\,Y(1) - Y(0) \mid X = x\,]$.
Under random assignment, estimate by the treated-minus-control difference *within the leaf*:

Let $L = L(x, T)$ and $\bar{Y}_{w,L}$ be the mean outcome among units in leaf L with $W = w$. Then

$$\hat{\tau}(x) = \bar{Y}_{1,L} - \bar{Y}_{0,L}.$$

Each leaf is treated as a small randomized experiment.

The tree reads off a different treatment effect in each leaf. But how do we *grow* it? The obvious objective is infeasible:

$$\min_T \sum_i \big(\tau_i - \hat{\tau}(X_i; T)\big)^2, \qquad \tau_i = Y_i(1) - Y_i(0) \ \text{never observed.}$$

How do we grow a causal tree?

We cannot chase τ_i. Instead, grow the tree to find splits that separate leaf treatment effects. The clean intuition is:

> prefer splits with large differences between $\hat{\tau}_L$ and $\hat{\tau}_R$,
>
> provided each child has enough observations and enough treatment variation.

Regression-tree analogy. A regression tree looks for splits that pull *leaf means* apart. A causal tree looks for splits that pull *leaf treatment effects* apart.

> Maximising apparent heterogeneity alone would favour noisy, tiny leaves. Honest causal trees therefore combine heterogeneity search with minimum leaf sizes, treatment-variation checks, and separate estimation samples.

Why we like causal trees

- Even though the target is $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$, the tree readily reveals heterogeneity: a different effect in each leaf, with the splits naming *which* covariates drive it.
- The revealed heterogeneity is not automatically "causal" — splits chosen on the data can manufacture spurious differences.
- The fix is honesty. Use one subsample to choose the splits and a *separate* subsample to estimate the leaf effects. Then the leaf-by-leaf estimates are valid and we can test their significance.

Bridge. Under random assignment the leaf estimator is the raw difference of means above. Once we residualise with Robinson, that same leaf regression becomes $\hat{\tau}_\ell = \sum_{i \in \ell} \tilde{W}_i \tilde{Y}_i / \sum_{i \in \ell} \tilde{W}_i^2$ — the local effect we will plug into the forest next.

Standard random forests as kernel smoothers

A regression tree partitions $\mathcal{X}$ into leaves $R_1, \ldots, R_L$ and predicts $\hat{\mu}(x) = \sum_\ell \bar{Y}_{R_\ell} \mathbf{1}[x \in R_\ell]$.

A forest averages B trees, each grown on a subsample. Key reformulation:

$$\hat{\mu}^{RF}(x) = \sum_{i=1}^{n} \alpha_i(x)\, Y_i, \qquad \alpha_i(x) = \frac{1}{B} \sum_{b=1}^{B} \frac{\mathbf{1}[x \in R_b(x_i)]}{|R_b(x_i)|}$$

- The weights $\alpha_i(x)$ sum to 1.
- They define a data-adaptive neighbourhood around the query point x.
- A random forest is a kernel smoother with an adaptive kernel.

Causal random forest: the core idea

Wager & Athey (2018): use forest weights to estimate a *local causal slope*, not a conditional mean.

Given forest weights $\alpha_i(x)$, run the Robinson regression locally:

$$(\hat{\tau}(x), \hat{\mu}(x)) = \arg\min_{\tau,\mu} \sum_{i=1}^{n} \alpha_i(x) \left(\tilde{Y}_i - \mu - \tau \tilde{W}_i \right)^2.$$

Define weighted residual means $\bar{\tilde{W}}_\alpha = \sum_i \alpha_i(x) \tilde{W}_i$ and $\bar{\tilde{Y}}_\alpha = \sum_i \alpha_i(x) \tilde{Y}_i$. The local slope is

$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x)(\tilde{W}_i - \bar{\tilde{W}}_\alpha)(\tilde{Y}_i - \bar{\tilde{Y}}_\alpha)}{\sum_i \alpha_i(x)(\tilde{W}_i - \bar{\tilde{W}}_\alpha)^2}$$

Recognise it: this is Frisch–Waugh–Lovell, local, with forest-defined weights. The forest defines the neighbourhood; Robinson defines the causal slope.

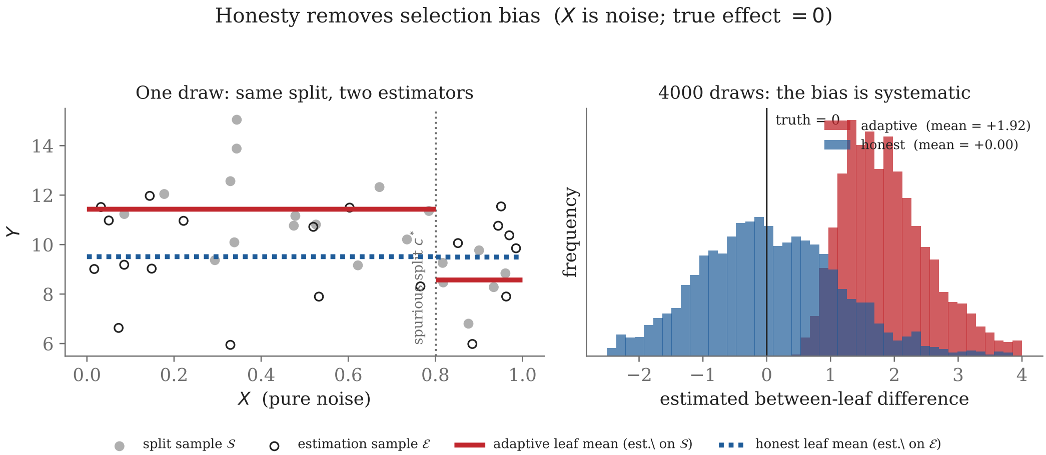
The problem with adaptive trees

- In a standard tree, splits are chosen after seeing the outcomes.
- So leaf values are selected on the outcome — biased estimators.
- For *prediction*, averaging across trees mitigates this. For *inference*, the bias does not vanish asymptotically.
- Without honesty, the standard Wager–Athey Gaussian asymptotics do not apply. The usual CRF confidence intervals are no longer justified.

Honest tree. For each training observation i, use i either to *determine the split structure or* to *estimate leaf values* but not both.

In practice: split the sample into $\mathcal{S}$ (splitting) and $\mathcal{E}$ (estimation), with $\mathcal{S} \cap \mathcal{E} = \emptyset$.

Honesty figure



The best linear projection (BLP) test

Run the regression:

$$Y_i = \alpha_0 + \alpha_1 \hat{\tau}(X_i)\,(W_i - \hat{e}(X_i)) + \beta\,(W_i - \hat{e}(X_i)) + \gamma(X_i) + \varepsilon_i$$

Read the diagnostics:

- $\alpha_1 \approx 1 \Rightarrow$ CRF is well-calibrated.
- Test $H_0 : \alpha_1 = 1 \Rightarrow$ specification check.
- $\alpha_1 > 0$ significantly $\Rightarrow$ rejects “no heterogeneity”.

Reading a CATE distribution

After fitting, `cf.effect(X)` gives a distribution of estimates. How to interpret it:

Key insight

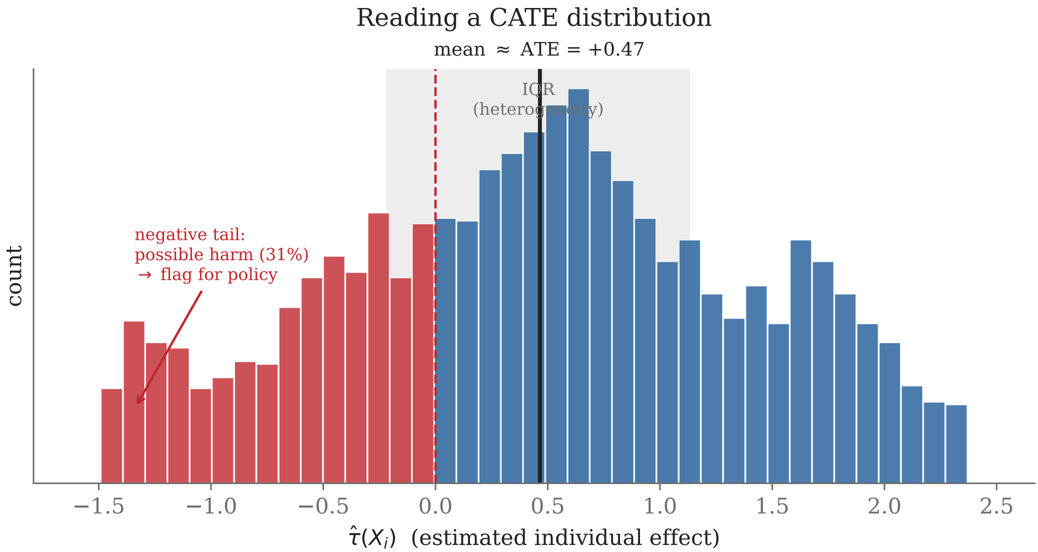
Mean $\approx$ ATE — same as classical estimators.

Spread (IQR) — degree of heterogeneity. A tight distribution around the ATE says heterogeneity is weak. A wide, skewed distribution says targeting matters.

Tails — who benefits most / least. Negative tail: possible harm for a subgroup — flag for policy.

Before trusting the spread: run the calibration (BLP) test from earlier (`test_calibration`), a wide histogram is necessary but not sufficient; $\hat{\alpha}_1$ significantly > 0 provides statistical evidence that the estimated heterogeneity is not just noise.

Cate distribution



Variable importance: what drives heterogeneity?

Forest-based variable importance (`cf.feature_importances_`): depth-weighted fraction of splits on each variable.

High importance $\Rightarrow$ the forest splits on this variable to detect heterogeneity.

Low importance $\Rightarrow$ the variable does not predict treatment-effect variation (even if it predicts Y strongly).

Key warning: feature importance $\neq$ effect size.

- High-importance variable: *heterogeneity changes with this covariate*.
- This is not the same as saying the subgroup with high X_j has a high τ.
- Always plot $\hat{\tau}(X_i)$ against the top-importance variables.

What does CRF find in the field?

Three questions we will answer:

Does meaningful treatment-effect heterogeneity exist?

Which covariates drive it?

What should a policymaker do with it?

Application 1 — Job training (National Job Corps Study)

Setting: Job Corps — randomised evaluation of the offer of job-training (intent-to-treat).

n	$\approx$ 15,400	participants (random assignment)
W	1	if assigned to the Job Corps offer
Y		weekly earnings (survey, 4 yrs; tax data, 9 yrs)
X		age, education, prior earnings, race, site

Key findings (Schochet, Burghardt & McConnell 2008):

- ATE is modest — about \$1,350/yr in Year 4 (≈\$26/week, ITT).
- Gains fade for the full sample over the 9-year horizon (tax data) . . .
- . . . except for the oldest participants (ages 20–24), whose earnings gains persist for years.
- Age heterogeneity is exactly the kind of structure a CRF surfaces *automatically* — without a pre-specified subgroup.

Application 2 — 401(k) eligibility and savings

Setting: 1991 SIPP — 401(k) eligibility as a conditional natural experiment (Poterba–Venti–Wise).

n	9,915	households
W	1	eligible for a 401(k) plan
Y		net financial assets
X		age, income, family size, education, marital status, pensions, IRA, home ownership

Key findings (Chernozhukov et al. 2018; Chernozhukov & Hansen 2004):

- ATE $\approx$ \$8,000–\$9,000 on net financial assets (DML) — the canonical DML benchmark.
- Identification rests on eligibility being exogenous *conditional on income* — the standard but contested assumption.
- Heterogeneity across the wealth distribution: quantile effects grow across net-asset quantiles.
- Economic debate: *forced saving* for liquidity-constrained households vs. *crowding-out* / asset-shifting for wealthier households.

Five lines of Python

```
# econml -- the Robinson/DML engine in Python
from econml.dml import CausalForestDML
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier

# Step 1:  fit (Robinson nuisances + honest forest, internally)
cf = CausalForestDML(
model_y=RandomForestRegressor(n_estimators=500),
model_t=RandomForestClassifier(n_estimators=500),
n_estimators=2000, discrete_treatment=True,
honest=True, random_state=0)
cf.fit(Y, W, X=X)

# Step 2:  CATE point estimates + 95% confidence intervals
tau_hat  = cf.effect(X)
lb, ub   = cf.effect_interval(X, alpha=0.05)

# Step 3:  average treatment effect
ate  = cf.ate(X)
```

Five lines. `CausalForestDML` wraps the full Robinson pipeline: cross-fit nuisances, honest forest, IJ variance. Open the notebook — we will run this live.

Key hyperparameters (CausalForestDML)

Parameter	Role	Practical guidance
`n_estimators`	Number of trees B	≥ 2000 for stable IJ variance; more is safer
`min_samples_leaf`	Minimum leaf size k	Default 10; raise if few treated per leaf
`max_samples`	Subsampling rate s/n	Default 0.45; lower $\Rightarrow$ more regularisation
`max_features`	Features per split	Default `"auto"` ($\approx \lceil p/3ceil$)
`honest`	Honest splitting	Always `True` — required for valid CIs
`cv`	Cross-fitting folds	Default 5; use 3 if $n < 1000$

Tip. Pass `RandomForestRegressor(n_estimators=500)` for nuisances (model_y, model_t). Gradient boosting also works; use cross-validated models for large n. The `cv=5` folds handle cross-fitting automatically.

Common pitfalls

- Overlap. Plot $\hat{e}(X_i)$ separately for treated and controls. Scores near 0 or 1 violate positivity — trim or use overlap-weighted ATE.
- Calibration. Use `RScorer` from `econml.score` or run the BLP manually: regress Y on $\hat{\tau}(X)\tilde{W}$. Slope $\approx 1 \Rightarrow$ calibrated; slope significantly $> 0 \Rightarrow$ real heterogeneity.
- Deconfounder check. If unconfoundedness is doubtful, use `econml.iv.dml.DMLIV` or run sensitivity analysis (Rosenbaum bounds).
- Bias/variance trade-off. Small leaves $\Rightarrow$ low bias, high variance. Large leaves $\Rightarrow$ low variance, may miss heterogeneity. Increase `min_samples_leaf` if CIs are too wide.

Connections to other methods

Method	Relationship to CRF
X-learner (Künzel et al.)	RF for imputed treatment effects; no asymptotic theory
R-learner (Nie & Wager)	Same Robinson decomposition; general ML for $\hat{\tau}$
DR-learner	AIPW score as pseudo-outcome; orthogonal to nuisance errors
BART (Chipman et al.)	Bayesian posterior; no frequentist CI guarantee
Causal BART	BART + BCF regularisation; slower but competitive
DML (Chernozhukov et al.)	ATE version of CRF; same orthogonality machinery

CRF sits at the intersection of the R-learner family and double machine learning — the non-parametric, asymptotically valid member.

The four innovations

Causal Random Forests give non-parametric, asymptotically valid inference on heterogeneous treatment effects under unconfoundedness.

Four key innovations:

1. The Robinson engine. Residualise Y, W by $\hat{m}, \hat{e}$; orthogonalise the causal signal from confounding.
2. Causal splitting. Trees split to maximise $\tilde{Y}/\tilde{W}$ covariance heterogeneity, not outcome variance.
3. Honesty. Decoupled splitting and estimation subsamples — enables Gaussian asymptotics.
4. IJ variance. Consistent variance estimates without refitting.

When to use CRF — and when not to

Use CRF when:

- You want *heterogeneous* effects — a function $\tau(x)$, not a single number.
- Unconfoundedness is plausible (good covariates, no obvious selection).
- You need confidence intervals, not just point estimates.
- n is in the thousands; p is moderate to high.

Don't use CRF when:

- Unconfoundedness clearly fails, use IV, RDD, or DiD instead.
- Overlap is poor everywhere, no comparison group exists.
- You only care about the ATE, use DML, it's simpler.
- n is small (a few hundred), the forest cannot adapt usefully.

Never assume CRF solves unobserved confounding. It doesn't.
It *exploits* unconfoundedness; it doesn't *create* it.

Key references I

- Wager, S. & Athey, S. (2018). Estimation and Inference of Heterogeneous Treatment Effects using Random Forests. *JASA*, 113(523).
- Wager, S., Hastie, T. & Efron, B. (2014). Confidence Intervals for Random Forests: The Jackknife and the Infinitesimal Jackknife. *JMLR*, 15.
- Athey, S., Tibshirani, J. & Wager, S. (2019). Generalized Random Forests. *Annals of Statistics*, 47(2).
- Athey, S. & Wager, S. (2019). Estimating Treatment Effects with Causal Forests: An Application. *Observational Studies*, 5(2), 37–51.
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1).
- Robinson, P.M. (1988). Root-N consistent semiparametric regression. *Econometrica*, 56(4).
- Rosenbaum, P. & Rubin, D. (1983). The central role of the propensity score. *Biometrika*, 70(1).
- Holland, P.W. (1986). Statistics and Causal Inference. *JASA*, 81(396).
- Nie, X. & Wager, S. (2021). Quasi-Oracle Estimation of Heterogeneous Treatment Effects. *Biometrika*, 108(2).
- Chernozhukov, V. et al. (2018). Double/Debiased Machine Learning. *The Econometrics Journal*, 21(1).

Key references II

- Yadlowsky, S. et al. (2021). Evaluating Treatment Prioritization Rules via Rank-Weighted Average Treatment Effects. *arXiv:2111.07966*.
- Künzel, S. et al. (2019). Metalearners for Estimating Heterogeneous Treatment Effects using Machine Learning. *PNAS*, 116(10).
- Schochet, P.Z., Burghardt, J. & McConnell, S. (2008). Does Job Corps Work? Impact Findings from the National Job Corps Study. *American Economic Review*, 98(5).

Thank you

Questions?

Remember

Residualise. Regress residual on residual. Locally. Honestly.

Backup — the Robinson derivation in full

1. **Conditional mean** (consistency + unconfoundedness):

$$\mathbb{E}[Y_i \mid X_i, W_i] = \mathbb{E}[Y_i(0) \mid X_i] + \tau(X_i)\, W_i.$$

2. **Average over $W \mid X$:**

$$m(X_i) = \mathbb{E}[Y_i(0) \mid X_i] + \tau(X_i)\, e(X_i) \;\Rightarrow\; \mathbb{E}[Y_i(0) \mid X_i] = m(X_i) - \tau(X_i)\, e(X_i).$$

3. **Substitute back:**

$$\mathbb{E}[Y_i \mid X_i, W_i] = m(X_i) + \tau(X_i)\left(W_i - e(X_i)\right).$$

4. **Add noise $\varepsilon_i = Y_i - \mathbb{E}[Y_i \mid X_i, W_i]$:**

$$Y_i = m(X_i) + \tau(X_i)\, \tilde{W}_i + \varepsilon_i \quad \text{(Robinson)}.$$

5. **Subtract $m(X_i)$:**

$$\tilde{Y}_i = \tau(X_i)\, \tilde{W}_i + \varepsilon_i.$$

By construction $\mathbb{E}[\varepsilon_i \mid X_i, W_i] = 0$ and $\mathbb{E}[\tilde{W}_i \mid X_i] = 0$.

Backup — closing the local regression of slide 20

Objective (two-parameter weighted least squares; weights $\sum_i \alpha_i(x) = 1$):

$$(\hat\tau(x), \hat\mu(x)) = \arg\min_{\tau,\mu} \sum_{i=1}^{n} \alpha_i(x)\left(\tilde Y_i - \mu - \tau\,\tilde W_i\right)^2.$$

First-order conditions.

$\partial/\partial\mu$:

$$\sum_i \alpha_i(x)\left(\tilde Y_i - \mu - \tau \tilde W_i\right) = 0.$$

Since $\sum_i \alpha_i(x) = 1$, this is $\bar{\tilde Y}_\alpha = \mu + \tau \bar{\tilde W}_\alpha$, i.e.

$\boxed{\hat\mu = \bar{\tilde Y}_\alpha - \tau\,\bar{\tilde W}_\alpha}$ (line through weighted means).

$\partial/\partial\tau$:

$$\sum_i \alpha_i(x)\,\tilde W_i\left(\tilde Y_i - \mu - \tau \tilde W_i\right) = 0.$$

Substitute $\hat\mu$ and collect τ:

$$\sum_i \alpha_i(x)\tilde W_i \tilde Y_i - \bar{\tilde W}_\alpha\,\bar{\tilde Y}_\alpha = \tau\left(\sum_i \alpha_i(x)\tilde W_i^2 - \bar{\tilde W}_\alpha^{\,2}\right).$$

Centering identity. Because $\sum_i \alpha_i(x) = 1$, both sides above are weighted (co)variances: $\sum_i \alpha_i(x)\tilde W_i \tilde Y_i - \bar{\tilde W}_\alpha\,\bar{\tilde Y}_\alpha = \sum_i \alpha_i(x)(\tilde W_i - \bar{\tilde W}_\alpha)(\tilde Y_i - \bar{\tilde Y}_\alpha)$, denominator analogous. Hence

$$\boxed{\hat\tau(x) = \frac{\sum_i \alpha_i(x)\,(\tilde W_i - \bar{\tilde W}_\alpha)(\tilde Y_i - \bar{\tilde Y}_\alpha)}{\sum_i \alpha_i(x)\,(\tilde W_i - \bar{\tilde W}_\alpha)^2}}$$

(weighted cov over weighted var — local FWL).

Eliminating μ centers the residuals to $(\tilde W_i - \bar{\tilde W}_\alpha)$; the forest enters only through the weights $\alpha_i(x)$.

Backup — Frisch–Waugh–Lovell in matrix form

Full model, W a single column, X the controls (intercept included):

$$Y = W\tau + X\beta + u.$$

Annihilator matrix. $M_X = I - X(X'X)^{-1}X' = I - P_X,$ symmetric $(M_X' = M_X)$, idempotent $(M_X^2 = M_X)$, and $M_X X = 0$.

Premultiply the model by M_X. The control block is killed, $M_X X\beta = 0$:

$$\underbrace{M_X Y}_{\tilde{Y}} = \underbrace{M_X W}_{\tilde{W}}\,\tau + \underbrace{M_X X\beta}_{=0} + M_X u \implies \tilde{Y} = \tilde{W}\,\tau + M_X u.$$

Residualising Y and W on X leaves the *same* τ.

OLS of $\tilde{Y}$ on $\tilde{W}$, using $M_X' M_X = M_X$:

$$\hat{\tau} = (\tilde{W}'\tilde{W})^{-1}\tilde{W}'\tilde{Y} = (W'M_X W)^{-1}W'M_X Y.$$

Scalar form (W univariate, residuals mean-zero):

$$\boxed{\hat{\tau} = \frac{\sum_i \tilde{W}_i\,\tilde{Y}_i}{\sum_i \tilde{W}_i^{\,2}}}$$

FWL theorem. This equals *exactly* the W-coefficient from the full multiple regression $Y = W\tau + X\beta + u$.

Partial out X first, then regress residual on residual — same τ.

Robinson (1988) generalisation. Replace the *linear* projection P_X by the *conditional expectation*: $P_X Y \to m(x) = \mathbb{E}[Y \mid X = x]$ and $P_X W \to e(x) = \mathbb{E}[W \mid X = x]$. Same partialling-out, but $g(X)$ may be arbitrarily nonlinear; the forest runs it *locally* with weights $\alpha_i(x)$ — FWL made local.

Backup — why $n^{-1/4}$ is enough

The orthogonal score removes the *first-order* nuisance error, so the bias becomes a **product** of two errors. That is the whole trick.

Naive plug-in (non-orthogonal score)

$$\text{Bias} = O\Big(\|\hat{m} - m\|\Big) \sim n^{-1/3}$$

Scale by $\sqrt{n}$ for inference:

$$\sqrt{n} \cdot n^{-1/3} = n^{\frac{1}{2} - \frac{1}{3}} = n^{1/6} \longrightarrow \infty$$

Bias dominates $\Rightarrow$ CIs invalid.

DML (Robinson / orthogonal score)

$$\text{Bias} = O\Big(\|\hat{m} - m\| \cdot \|\hat{e} - e\|\Big)$$

Each nuisance only $o(n^{-1/4})$:

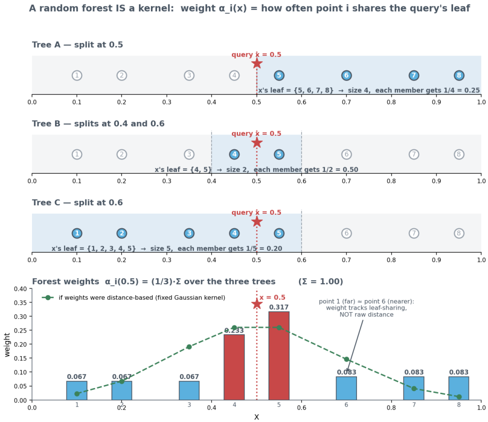
$$o\Big(n^{-1/4}\Big) \cdot o\Big(n^{-1/4}\Big) = o\Big(n^{-1/2}\Big)$$

$$\sqrt{n} \cdot o\Big(n^{-1/2}\Big) = o(1) \longrightarrow 0$$

Bias vanishes $\Rightarrow$ clean $\mathcal{N}(0, 1)$.

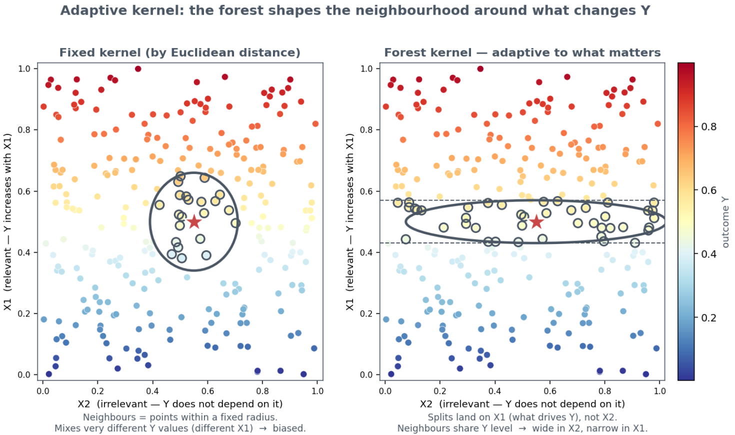
- **Two numbers, two roles.** $1/2$ is the *target* rate for $\hat{\tau}$ — parametric, the speed that buys valid inference. $1/4$ is all you must demand of *each* nuisance, because two $n^{-1/4}$ errors *multiply* into one $n^{-1/2}$.
- **Why a product.** Neyman orthogonality zeroes the linear term in the bias expansion (this is exactly $\mathbb{E}[\widetilde{W} \mid X] = 0$); the first surviving term is the cross term $\delta_m \cdot \delta_e$.
- **Cheap deal.** Neither $\hat{m}$ nor $\hat{e}$ need parametric accuracy — a forest or lasso beating $n^{-1/4}$ under regularity is enough.

Backup — the forest as a kernel (worked example)



Read it: top strips = the same 8 points seen by 3 trees; *blue* points share the query's leaf, *grey* do not. The weight $\alpha_i(0.5)$ (bottom) just averages each tree's $1/|\text{leaf}|$ contribution. Weights sum to 1. The green dashed line is what a fixed distance kernel would give — the forest disagrees: weight tracks *leaf-sharing*, not raw distance.

Backup — why the kernel is adaptive



Y depends only on $X1$: the fixed circle (left) mixes very different Y; the forest's band (right) is wide along the irrelevant axis, narrow along the relevant one — a *learned* kernel.

Backup — wage example: why the T-learner fails

T-learner failure mode

Separate outcome models can fit Y well while cancelling or amplifying noise in $\hat{m}_1(x) - \hat{m}_0(x)$.

Estimator	Estimate	
True effect	**2,000**	flat, no heterogeneity
Naive difference in means	18,583	confounded by age (treated are older)
owcolorrowhi T-learner $\hat{\mu}_1 - \hat{\mu}_0$	5,274	swings 665–12,988: spurious heterogeneity
DML / Robinson	1,888	$\approx$ truth

Double burden: each model must fit the steep age–wage curve *and* the effect; where a group is thin it extrapolates, and baseline errors leak into $\hat{\mu}_1 - \hat{\mu}_0$. Robinson cancels $g(\text{age})$ first, leaving the flat 2,000.